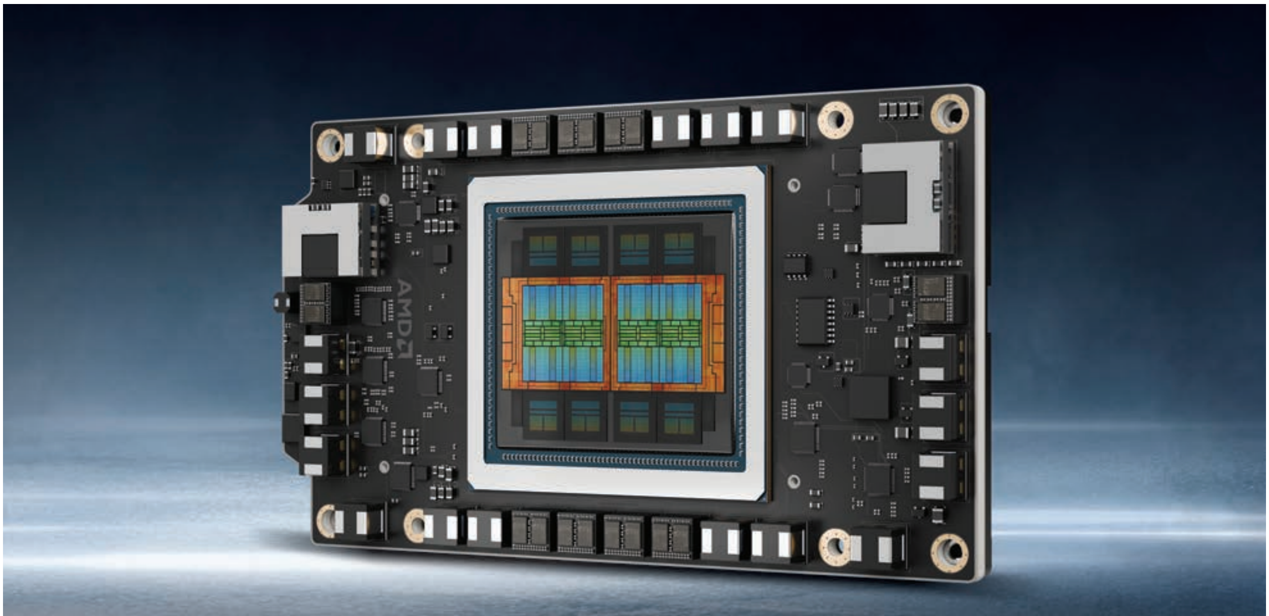

AMD Instinct MI350:

A deep dive into Team Red's AI powerhouse

A deep dive into AMD's Instinct MI350 GPU—featuring CDNA 4, 288GB HBM3E, and 3D chiplets—designed to take on NVIDIA in the AI accelerator race.

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The AI accelerator landscape is undergoing one of its most competitive phases in years. AMD's newly announced Instinct MI350 series arrives as a bold response to NVIDIA's Blackwell architecture and the growing computational demands of generative AI and large-scale language models. With 185 billion transistors, support for up to 288 GB of HBM3E memory, and the new CDNA 4 architecture, AMD has made it clear that it's ready to compete toe-to-toe in the data centre.

Compared to the MI325, the MI350 is a full architectural rethink. The CDNA 4 architecture introduces a chiplet-based approach with an emphasis on low-precision AI workloads, high bandwidth memory, and improved scalability. In a market increasingly defined by large model sizes and inference efficiency, AMD's design philosophy is finally aligned with the future of AI.

CDNA 4 architecture: The heart of MI350

CDNA 4 represents a significant evolution from CDNA 3, which debuted with the MI300X. While the previous generation made strides in compute density and mixed-precision performance, CDNA 4 focuses sharply on the needs of modern AI. Central to the MI350X is a new 3D chiplet design that stacks eight compute dies, referred to as XCDs, on top of just two I/O dies, or IODs. This is a departure from the four-IOD layout in the MI300 series.

Reducing the number of IODs is more than a simplification. It shortens the physical distance data must travel,